

FedCL: Detecting Backdoor Attacks in Federated Learning with Confidence Levels

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Abstract—Federated Learning (FL) enables multiple clients to collaborate in training neural network models while retaining their private data locally. Despite its advantages, FL is vulnerable to backdoor attacks due to its distributed nature. Attackers introduce triggers into the global model, causing it to make specified predictions on inputs containing these triggers. Existing detection or clustering defense methods based on distance and similarity have significant limitations. Methods based on clipping and adding noise can only slightly mitigate the impact of backdoor attacks. To achieve a better defense, we introduce FedCL, a backdoor defense framework that accurately detects backdoor models by assessing the uncertainty of model predictions. Additionally, it employs dynamic clipping to limit model updates' impact, successfully mitigating backdoor attacks without compromising the global model's accuracy. The experiments indicate that FedCL moderately improves by 0.01% $\uparrow$ ~85.78% $\uparrow$ than the state-of-the-art (SOTA) defense methods, especially in the CIFAR-10 task trained with more complex networks.

Index Terms—Federated learning, backdoor defense, backdoor detection, distributed, uncertainty

I. INTRODUCTION

Federated Learning (FL) [1] enables multiple clients to train a promising global model with neural network models collaboratively. It is a positive outcome of FL that various applications are promoted, including image processing [2], natural language processing [3], and etc., while it also opens up the possibility for adversaries to effortlessly manipulate datasets and upload models to inject triggers in targeted attacks. More specifically, adversaries posing as benign clients inject distinct triggers into a tiny portion of the training datasets to trick the trained model into being strongly associated with these triggers. Additionally, adversaries cannot severely affect the performance of the injected models on other benign datasets but rather boost the performance of the injected models on malicious datasets. For backdoor attacks

in practical applications, FL has proven to be more vulnerable to security threats.

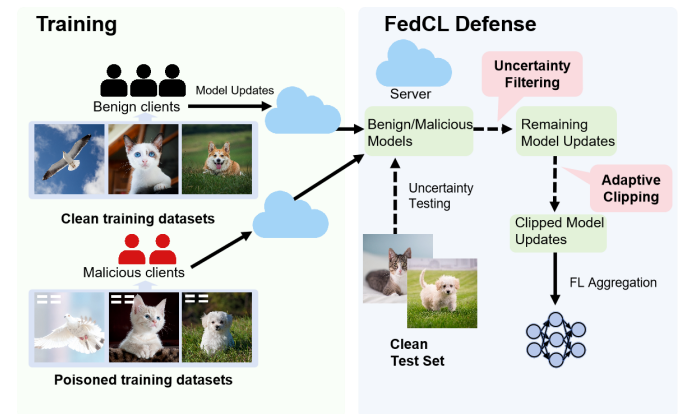


Fig. 1. Overview of FedCL

Although there have been intensive studies designing robust aggregation methods and empirical robust federated training protocols against backdoor attacks, existing approaches are ineffectively eliminating the backdoor. More generally, the existing defense methods in FL involving robust aggregation strategies [4], [5] that detect and reject malicious model updates and pre-process defense strategies [6]–[8] that limit and weaken the impact of malicious model updates. For robust aggregation, these methods primarily compare the distance and similarity between individual local models or between local and global models. However, detection methods have significant limitations in distinguishing the distances between malicious and benign models. More generally, these distance-based strategies can quickly fail in scenarios with non-IID data distribution, as each client's local model exhibits significant variations. Additionally, adversaries can evade such defenses by constraining the distance or similarity of model updates. For pre-process defense, clipping and adding noise to the model submitted by clients. However, this approach only slightly

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mitigates the poisoning effects and has a certain negative impact on the global model's accuracy.

To bridge this gap, we propose the Detecting Backdoor Attacks in Federated Learning Defense Framework (FedCL) with confidence levels, as shown in Fig.1, which enables to detect backdoor model updates in FL with high accuracy. For detecting backdoor attacks, we propose uncertainty filtering to distinguish the differences between benign models and malicious ones through uncertainty. More generally, we perform uncertainty testing on the models submitted by clients using a small collection of clean test sets collected by the server each round. Before aggregating model updates, we utilize dynamic clipping to limit all model updates for reducing the potential impact of undetected malicious model updates on the accuracy of the global model. Our contributions are three-folded:

- In this work, we propose Detecting Backdoor Attacks in Federated Learning Defense Framework (FedCL) with confidence levels, which fully defends against adversaries' backdoor attacks without a strong assumption about the adversary's attack strategy and/or the underlying distribution of benign or adversarial datasets.
- Analyzing backdoor attacks in FL, we propose a novel detection method to mitigate them. More generally, for filtering malicious models, we assess the prediction uncertainty between malicious models and benign ones. Furthermore, the detection method can perfectly identify malicious models, significantly outperforming existing detection and defense approaches.
- We empirically evaluate the effectiveness of FedCL on benchmark datasets compared with SOTA defense methods. Furthermore, we conducted experiments under conditions that could potentially impact the effectiveness of FedCL. The results demonstrate that FedCL outperforms SOTA algorithms significantly regarding defense effectiveness.

II. RELATED WORK

A. Federated Learning

Federated Learning [1] is capable of collaboratively training deep learning models by multiple clients under the supervision of a centralized aggregator. To reduce the communication overhead, the server in FedAvg [1] only needs to select a subset of clients to participate in training and aggregation through LocalSGD. More generally, for m clients, the aggregation in FedAvg can be formulated as:

$$\mathcal{F}_i(w) = \frac{1}{|D_i|} \sum_{j \in D_i} \ell(w, \{x_j, y_j\}) \quad (1)$$

where $\mathcal{D}_i$ is a local dataset holding by client- i , $\forall i \in [m]$, w is the local model, and $\mathcal{F}_i(w)$ evaluates the local empirical risk over $\{x_j, y_j\} \in \mathcal{D}_i$ of client- i . To ensure robustness in the aggregation process, several Byzantine-Robust aggregation rules have been proposed, such as Krum [5], Trimmed Mean [9], and Median [9]. It is a positive outcome of the attractive advantages of FL while it also opens up the possibility for

adversaries to effortlessly manipulate datasets and upload models to inject triggers in targeted attacks [6], [10],

B. Backdoor Attacks on Federated Learning

An adversarial trigger is inserted into training data to corrupt it, resulting in incorrect predictions on test data with the same triggers. More generally, backdoor attacks can be categorized into data poisoning and model poisoning, which are described as follows:

- **Data Poisoning.** For data poisoning, the poisoned training dataset includes both clean samples and poisoned samples with a backdoor trigger. The proportion of poisoned data to all data is referred to as Poisoned Data Rate (PDR). To balance the effectiveness and stealth of the attack, the adversary chooses an appropriate PDR for training a malicious model.
- **Model Poisoning.** Due to the data-poisoned models are susceptible to being detected, combining data poisoning with model poisoning techniques can enhance the effectiveness of backdoor attacks. Attackers can control the training process and hyperparameters (e.g., epochs and learning rate) and freely modify model parameters before submitting the model [6]. Some representative model poisoning attacks include the projected gradient descent (PGD) attack with model replacement [11], the constrain-and-scale attack [6].

III. PROBLEM FORMULATION

A. Problem Definition

To escape detection, malicious attackers aim to manipulate their local updates to inject backdoor triggers into the global model. Specifically, these triggers are designed so that the global model will provide predetermined target predictions when the trigger pattern appears during the inference stage. To avoid detection by the defense method, malicious attackers strive to maintain regular prediction accuracy on clean tasks. For $\forall \{x, y\} \in \mathcal{D}$, the poisoned model has the outputs below:

$$G'(x) = \begin{cases} G(x, y), & \forall \{x, y\} \notin \mathcal{D}_p \\ G(x, y_{target}), & \forall \{x, y_{target}\} \in \mathcal{D}_p \end{cases}, \quad (2)$$

where G denotes the global model, G' denotes the local poisoned model, $\mathcal{D}_p$ denotes the poisoned dataset belonging to the adversary, and y_{target} denotes the specific label modified by the adversary of the input x .

B. Backdoor Attacks and Defense

Considering previous work [7], [12], we assume that the adversary can control a maximum of 50% of all clients. Moreover, a potential adversary is aware of the aggregation rules and defense mechanisms employed by the aggregator. For backdoor attacker, the adversary can make arbitrary modifications to the models submitted for the final aggregation, and manipulate the local training process, including training parameters. In addition, the adversary has no control over any processes or parameters executed within the aggregator. For the server, it cannot access the raw local training data on

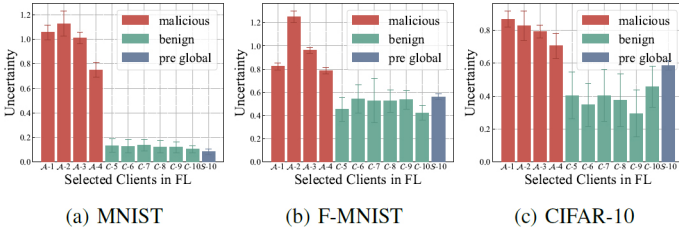


Fig. 2. Uncertainty over different benchmark datasets in FL with attackers (A), clients (C), and server (S).

the clients or control the clients' local training processes but fully access the global model and local model updates from all clients. Furthermore, the server can collect a small amount of clean testing data. In contrast to externally available poisoned datasets, the service provider can internally gather a clean test set at a minimal cost since only a few samples are needed (i.e., 100). For instance, Gao et al. [13] utilize a small amount of clean dataset collected by the server as their root dataset.

IV. FEDCL: DETECTING BACKDOOR ATTACKS IN FEDERATED LEARNING

A. Overview

FedCL aims to provide a series of defenses to address a typical scenario of backdoor attacks. In particular, the designed FedCL consists of two main steps as illustrated in Fig.1 and Algorithm 1. The procedure is summarized as follows: (1) *Uncertainty Filtering*: For aggregation, the aggregator performs uncertainty tests on each local model to identify and remove malicious ones. (2) *Adaptive Clipping*: The selected model updates are clipped with a dynamic threshold.

Algorithm 1 Detecting Backdoor Attacks in Federated Learning Defense Framework (FedCL)

Require: Initial model parameters G_0 , total training round T , randomly select a set of clients K , $|inds|$ is the accepted number of clients by the aggregator, S is the median of all model update norms, and α is the coefficient controlling the clipping threshold.

Ensure: Global model G_T at the T rounds.

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1: for each training iteration  $t$  in  $[1, T]$  do
2:   for each client  $i$  in  $[1, K]$  do
3:      $\mathcal{W}_i \leftarrow \text{CLIENTUPDATE}(G_{t-1})$ 
4:     Send  $\mathcal{W}_i$  to the Server.
5:   end for
6:   # Uncertainty-based Filtering
7:    $inds = \text{UNCERTAINTYFILTER}(\mathcal{W}_1, \dots, \mathcal{W}_K)$ 
8:   # Adaptive Clipping
9:    $S = \text{MEDIAN}(\|\mathcal{W}_1\|_2, \dots, \|\mathcal{W}_K\|_2)$ 
10:   $S_t = \alpha \times S$ 
11:  for each client  $i$  in  $inds$  do
12:     $C_i = \mathcal{W}_i \times 1/\max(1, \|\mathcal{W}_i\|_2 \times 1/S_t)$ 
13:  end for
14:   $G_t = \frac{1}{|inds|} \cdot \sum_{i=1}^{|inds|} C_i$ 
15: end for

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B. Uncertainty Filtering

Since the data distributions are extremely unbalanced among clients, existing defense methods [7] cannot distinguish benign models from malicious ones using clustering. Furthermore, for classification tasks in FL, the penultimate layer (PLR) demonstrates the extraordinary ability to distinguish malicious models from benign ones [14]. In particular, PLR is the last layer or the last layer above softmax on a neural network. However, it is extremely difficult to discriminate the backdoor attackers through PLR with different non-IID data.

In order to separate the malicious models and benign ones, we propose detecting malicious models through uncertainty filtering (line 7 in Algorithm 1). More generally, the trusted dataset ($\mathcal{D}_{clean}$) consists of a small amount of sample data (i.e., 100 samples), which is used as a vehicle to detect malicious models based on UNCERTAINTY (line 8 in Algorithm 2). To capture the uncertainty, we extract the features of PLR of each class from model inference with $\mathcal{D}_{clean}$, and then computes a probability via softmax and entropy, which is as follows:

$$\text{UNCERTAINTY}(\mathcal{W}) = - \sum_i \frac{e^{w_i^p}}{\sum_{c=1}^C e^{w_c^p}} \log \frac{e^{w_i^p}}{\sum_{c=1}^C e^{w_c^p}}, \quad (3)$$

where

$$w_i^p = \text{PLR}(\mathcal{W}, \mathcal{D}_{clean}), \quad (4)$$

$\mathcal{W}$ is the features from model inference, and w_i^p is the PLR features of $\mathcal{W}$ in each class i . Further, $\text{UNCERTAINTY}(\mathcal{W})$ corresponds to the uncertainty of the model inference for $\mathcal{D}_{clean}$ (line 1 and 7 in Algorithm 2). The higher the entropy, the higher the uncertainty, and the lower the confidence. We compared the uncertainty between benign models and malicious ones, as shown in Fig.2. It has been observed that the uncertainty of the malicious model is significantly higher than that of the benign model. Due to the poisoned samples with trigger mechanisms for local training, the model's confidence may decrease and uncertainty may increase in clean sample sets without triggers. Moreover, we set an uncertainty threshold to filter out high-uncertainty malicious models. During convergence, the uncertainty of the global model changes dynamically, making it better to determine this threshold based on its uncertainty. More detailed, the uncertainty filtering is shown in Algorithm 2.

C. Adaptive Clipping

In addition, clipping model updates from each client after uncertainty filtering can mitigate the impact of a relatively small number of malicious model updates that bypass uncertainty filtering algorithms and extreme-scale benign updates. In the process of convergence of the global model, the L2-norm of benign client model updates gradually decrease as the global model converges. For each round, the clipping threshold is applied dynamically to minimize the effect on benign model updates. Specifically, we first get the median of the L2-norms of all model updates (including those that have been filtered out). To eliminate the impact of escaping malicious model updates, we set α times the median as the

Algorithm 2 UncertaintyFilter

Require: Global model parameters G_{t-1} , all clients' updates $\mathcal{W}_1, \dots, \mathcal{W}_K$, global test set $\mathcal{D}_{clean}$, and p is the parameter for adjusting the threshold.

Ensure: Retained clients inds

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1:  $G = \text{MODELINFERENCE}(G_{t-1}, \mathcal{D}_{clean}, \text{PLR})$ 
2:  $S_g = \text{UNCERTAINTY}(G)$ 
3:  $S_t = S_g + p$ 
4: Initialize an array inds = []
5: for each client  $i$  in  $[1, K]$  do
6:    $\mathcal{W}_i \leftarrow G_{t-1} + \mathcal{W}_i$ 
7:    $\mathcal{W} = \text{MODELINFERENCE}(\mathcal{W}_i, \mathcal{D}_{clean}, \text{PLR})$ 
8:    $S_i = \text{UNCERTAINTY}(\mathcal{W})$ 
9: end for
10:  $S_m = \text{MEDIAN}(S_1, \dots, S_K)$ 
11: if  $S_t < S_m$  then
12:    $S_t = S_m$ 
13: end if
14: for  $i$  in  $[1, K]$  do
15:   if  $S_i \leq S_t$  then
16:     inds.append( $i$ )
17:   end if
18: end for

```

final clipping threshold (line 9~10 in Algorithm 1). We found that most benign model updates have L2-norms within the specified clipping threshold, ensuring the benign aggregation of the model. More detailed, the clipping process (line 12 in Algorithm 1) is as follows:

$$C_i = \mathcal{W}_i \times 1/\max(1, \|\mathcal{W}_i\|_2 \times 1/S_t), \quad (5)$$

where $\mathcal{W}_i$ is the model update of i -client, $\|\cdot\|_2$ is L2-norm, S_t is the clipping threshold, and C_i is the clipped model update.

V. EXPERIMENTAL EVALUATION

We implement a prototype of FedCL by PyTorch. Extensive experiments are conducted on the server equipped with 64-core CPUs, 128GB RAM, and 1 NVIDIA GeForce RTX 3060Ti. For continuous attack scenarios, we evaluate the performance of the designed FedCL defense scheme against the advanced distributed backdoor attack (DBA) [10] and compare it with 11 SOTA defense methods. These SOTA methods include the previously mentioned effective defense against backdoor attacks: Krum [5], Median [9], Bulyan Krum [15], Trimmed Mean [9], FoolsGold [16], RFA [17], DeepSight [18], RobustLR [19], Fltrust [13], Flame [7] and Fedward [12].

Dataset and models. We evaluate FedCL's performance on three commonly used benchmark datasets: MNIST, Fashion-MNIST (F-MNIST), and CIFAR-10. We employ a CNN network with two convolutional layers and two fully connected layers for MNIST and F-MNIST datasets, and the ResNet-18 architecture for the CIFAR-10 dataset.

Parameters setting. To ensure fairness, the experimental settings are consistent with those used in prior works. Specifically, we split the training data for FL clients in a non-

TABLE I
MA (%) AND BA (%) COMPARISON OF FEDCL WITH SOTA DEFENSES AGAINST DBA ATTACKS.

Defenses	MNIST		F-MNIST		CIFAR-10	
	MA	BA	MA	BA	MA	BA
Benign Setting	99.00	0.16	82.71	7.02	78.85	6.72
No Defense	98.88	100.00	81.60	99.33	78.27	86.58
Krum	97.08	0.94	77.60	16.23	75.69	97.48
Median	98.26	98.55	78.41	17.89	66.05	88.53
Bulyan Krum	98.50	0.27	79.16	14.26	76.24	96.96
Trimmed Mean	98.47	0.25	78.94	17.70	77.74	92.10
FoolsGold	98.71	100.00	81.10	99.11	77.97	85.93
RFA	98.76	100.00	81.74	61.23	78.84	40.91
DeepSight	98.84	100.00	81.67	99.31	78.29	86.72
RobustLR	98.69	100.00	79.62	55.84	77.23	87.46
Fltrust	98.66	0.13	82.24	14.03	77.30	53.19
Flame	98.92	89.55	79.68	24.57	76.71	93.17
Fedward	98.93	0.23	82.73	8.57	77.90	91.67
FedCL	98.96	0.12	82.18	6.58	79.84	5.89

IID manner using a Dirichlet distribution with the parameter $\alpha = 0.5$, which is more aligned with real-world application scenarios. Furthermore, in each round, we randomly choose 10 clients (i.e., 4 malicious and 6 benign clients) from a pool of 100 clients. The parameter p is set to 0.2, 0, and 0 for the MNIST, F-MNIST, and CIFAR-10 datasets, respectively, to fine-tune the uncertainty threshold. Additionally, by default, we randomly select 10 samples from each class in the corresponding test set of each dataset, totaling 100 samples as the server-side test set, and the clipping threshold α is set to 1.5.

Evaluation metrics. We assess the effectiveness of defense techniques against backdoor attacks using the following metrics: (1) *Backdoor Accuracy (BA)*: BA represents the model's accuracy in the backdoor tasks. (2) *Main Task Accuracy (MA)*: MA represents the model's accuracy in the main tasks. (3) *True Positive Rate (TPR)*: TPR represents the ratio of correctly identified poisoned models to the total number of models identified as poisoned. (4) *True Negative Rate (TNR)*: TNR represents the ratio of correctly identified benign models to the total number of benign models.

A. Comparison With SOTA

Effectiveness of FedCL against DBA attacks. Table I presents the evaluation results compared with SOTA defense mechanisms against DBA attacks and demonstrates the superiority of FedCL. We also evaluate the standard FL scheme in a purely benign setting and the FedAvg scheme without any defense. It can be observed that FedCL outperforms all evaluated SOTA defense methods, with a notable MA improvement by 1.27% ~ 13.79% $\uparrow$ on CIFAR-10. Additionally, FedCL effectively eliminates distributed backdoor attacks on all datasets (BA < 10%) without compromising the accuracy of the main task. It is worth noting that on CIFAR-10, FedCL achieves a higher MA than the Benign Setting. This is attributed to our dynamic clipping, which restricts the impact of extremely benign updates, enabling better model convergence. Some Byzantine-robust aggregation strategies (Krum, Median, Bulyan Krum, Trimmed Mean) successfully defend against DBA attacks on MNIST and F-MNIST but

TABLE II
MA/BA/TPR/TNR (%) COMPARISON OF FEDCL WITH SOTA DEFENSE METHODS UNDER VARIOUS PDRs.

PDR	Defenses	MNIST				F-MNIST				CIFAR-10			
		MA	BA	TPR	TNR	MA	BA	TPR	TNR	MA	BA	TPR	TNR
5/64	Flame	98.81	98.55	68.33	6.85	80.26	32.03	33.33	38.50	76.71	93.17	9.00	44.17
	Fedward	98.93	0.22	100.00	0.00	82.15	16.72	25.50	18.00	77.90	91.67	37.75	30.17
	Fltrust	98.64	0.13	-	-	80.48	68.61	-	-	77.30	53.19	-	-
	FedCL	98.97	0.11	100.00	0.37	81.17	12.32	71.50	29.83	79.84	5.89	99.50	4.50
10/64	Flame	98.92	79.53	74.44	2.78	79.81	41.07	48.75	29.00	76.11	90.80	13.75	41.17
	Fedward	98.94	0.24	100.00	0.00	82.69	11.29	57.25	8.33	77.94	85.10	45.00	25.83
	Fltrust	98.61	0.09	-	-	80.63	36.70	-	-	76.48	31.03	-	-
	FedCL	98.98	0.12	100.00	0.37	81.86	7.29	93.75	18.00	79.71	4.64	100.00	6.17
20/64	Flame	98.92	89.55	83.05	0.74	79.68	24.57	61.50	21.83	72.49	82.58	8.25	46.33
	Fedward	98.93	0.23	100.00	0.00	82.73	8.57	95.50	0.33	75.17	80.47	38.75	29.33
	Fltrust	98.66	0.13	-	-	82.24	14.03	-	-	75.44	9.32	-	-
	FedCL	98.96	0.12	100.00	0.37	82.18	6.58	100.00	13.83	79.92	4.76	100.00	6.67
32/64	Flame	98.91	0.15	99.72	0.19	81.07	20.63	71.00	17.50	72.11	82.06	5.25	48.17
	Fedward	98.93	0.25	100.00	0.00	82.49	6.96	99.75	0.17	72.91	77.42	28.00	30.17
	Fltrust	98.56	0.11	-	-	83.22	4.86	-	-	74.77	2.08	-	-
	FedCL	98.98	0.15	100.00	0.00	81.91	6.58	100.00	13.83	79.84	4.47	100.00	5.17

adversely affect the accuracy of the main task. On the more complex CIFAR-10 dataset, these methods completely fail as they mistakenly exclude benign updates. This exclusion amplifies the impact of malicious updates, leading to higher BA (97.48%, 88.53%, 96.96%, 92.10%) compared to the scheme without any defense (86.58%). FoolsGold, RFA, DeepSight, and RobustLR also have little to no defense effectiveness, possibly due to significant differences in data distribution between adversaries, making accurate adversary identification challenging. Fltrust performed well on the MNIST and F-MNIST datasets but failed on the CIFAR-10 dataset. Flame and Fedward also fail on the CIFAR-10 dataset because their clustering algorithms cannot accurately filter out malicious models. In summary, some SOTA defense methods mitigate DBA attacks on simpler MNIST and F-MNIST datasets but fail when confronted with the more complex CIFAR-10 dataset. In contrast, our FedCL successfully defends against DBA attacks with 5.89% BA.

Different poisoned data rates (PDR). Setting the PDR to 7.8125% (5/64), 15.625% (10/64), 31.25% (20/64), and 50% (32/64), we compare FedCL with SOTA defense methods, including Fltrust [13], Flame [7], and Fedward [12]. In this comparison, both Fltrust and FedCL utilize clean datasets on the server side. Flame and Fedward, similar to FedCL, prioritize the crucial step of eliminating malicious models to the best extent possible. In this subsection, we employ four metrics: MA, BA, TPR, and TNR, to demonstrate the advancement and robustness of FedCL. The comparison results are presented in Table II, with the lowest BA (i.e., best defense) bolded. On the MNIST dataset, Flame fails when PDR is below 50%, while the other three methods exhibit robust defense across different PDR settings. On the F-MNIST dataset, the performance of Fltrust is significantly influenced by the PDR setting. As the PDR decreases, Fltrust gradually loses its defensive capabilities with PDR = 7.8125%. FedCL and Fedward both demonstrate strong performance, with FedCL exhibiting superior BA and TPR, while Fedward shows better MA and TNR. On the CIFAR-10 dataset, Flame

and Fedward exhibit consistently low TPR across all PDR settings, suggesting the inefficacy of their clustering methods. The performance of Fltrust depends on the adversary's PDR attack strategy, and also fails under smaller PDR settings. In contrast, FedCL achieves nearly 100% TPR across all settings, demonstrating the effectiveness and robustness of FedCL's uncertainty-based detection method.

B. Performance Evaluation of FedCL

Performance with various test set sizes. We randomly choose n samples from each class in the respective test set of each dataset, where $n = 10, 5$, and 8 , respectively. This results in a total of $10 * n$ samples for the server-side test set. The performance of FedCL's defense effectiveness is reported in Table III, using metrics such as BA and TPR. We can observe that the defense effectiveness of FedCL improves with the increase in the size of the test set across different datasets. The impact of the test set size on the MNIST dataset is relatively small, where BA shows a slight increase from 0.12% to 0.24% as the test set size decreases. As the complexity of the data increases, the influence of the test set size on the defense effectiveness of FedCL gradually becomes more pronounced. When the test set size grows from 50 to 100, BA demonstrates a more significant decrease from 19.90% to 5.89%.

Performance with various data distributions. We employ a Dirichlet distribution with a hyperparameter α to simulate from non-IID to IID scenarios. We evaluate the effectiveness of FedCL with α set to 0.5, 2.0, and 5.0, as well as the IID scenario. In Table IV indicates that FedCL exhibits strong

TABLE III
BA (%) AND TPR (%) PERFORMANCE OF FEDCL ON VARIOUS TEST SET SIZES AT THE SERVER SIDE.

SIZE	MNIST		F-MNIST		CIFAR-10	
	BA	TPR	BA	TPR	BA	TPR
100	0.12	100.00	6.58	100.00	5.89	99.50
80	0.13	100.00	8.20	99.75	9.02	97.50
50	0.24	100.00	8.26	99.50	19.90	92.75

TABLE IV
MA (%) AND BA (%) PERFORMANCE OF FEDCL FROM IID TO NON-IID
WITH VARIOUS DEGREES.

α	MNIST		F-MNIST		CIFAR-10	
	MA	BA	MA	BA	MA	BA
0.5	98.96	0.12	82.18	6.58	79.84	5.89
2.0	98.82	0.19	83.70	6.28	79.95	5.76
5.0	99.03	0.18	83.63	5.93	79.59	6.34
IID	99.10	0.13	84.30	4.70	79.78	5.68

robustness in defending against DBA attacks across different α settings and the IID scenario. For instance, both MA and BA of the F-MNIST dataset show slight variations when transitioning from various non-IID to IID scenarios, ranging from 82.18% to 84.30% for MA and from 6.58% to 4.70% for BA.

VI. CONCLUSION

In this paper, we propose FedCL, a defense framework with confidence levels for detecting backdoor attacks in FL. Unlike previous defense methods relying on distance or similarity for detection or clustering, FedCL employs a small set of clean test samples to assess the prediction uncertainty of individual local models. It filters out malicious updates based on this uncertainty, combined with dynamic clipping to limit the impact of model updates. FedCL effectively mitigates backdoor attacks in FL while maintaining the optimal performance of the global model. Experimental results present that FedCL outperforms SOTA defense works, demonstrating efficiency and robustness.

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